Merkhilfe matplotlib

ktheu.github.io/python-uebungen

1 Grundlagen

Start

matplotlib_grundlagen

```
import matplotlib.pyplot as plt
import numpy as np
x = np.linspace(-3, 3, 200)  np.arange(0, 10, 0.5)
plt.plot(x, np.sin(x))
plt.show()
```

Farben, Linien, Marker

```
plt.plot(x, y, 'ro--')                          Kurzform
color='red'  color='C1'  '#1f77b4'
linestyle='--'  ':'  '-.'
linewidth=2  marker='o'  's' '^' 'x'
markersize=8  markerfacecolor='white'
```

Beschriften

```
plt.title('Titel')
plt.xlabel('x')  plt.ylabel('y')
plt.plot(x, y, label='sin')
plt.legend()  plt.legend(loc='upper left')
plt.grid(True)  plt.grid(axis='y')
```

Achsen einstellen

```
plt.xlim(0, 10)  plt.ylim(-1, 1)
plt.xticks([0, 1, 2])
plt.axhline(0, color='black', linewidth=1)
plt.axvline(2, linestyle='--')
plt.axis('equal')  plt.axis('off')
```

Text, Punkte, Pfeile

```
plt.plot(2, 4, 'o')                             einzelner Punkt
plt.text(2, 4, 'P', ha='center', fontsize=12)
plt.annotate('Hochpunkt', xy=(1, 2),
  xytext=(2, 3), arrowprops=dict(arrowstyle='->'))
```

Größe und Speichern

```
plt.figure(figsize=(8, 4))                      Zoll
plt.savefig('bild.png', dpi=150)                vor show()
```

2 Diagrammtypen

Streudiagramme

matplotlib_diagrammtypen

```
plt.scatter(x, y, s=40, color='C0', alpha=0.5)
marker='s'  edgecolor='black'
plt.scatter(x, y, c=werte, cmap='viridis')
plt.colorbar()
m, b = np.polyfit(x, y, 1)                      Ausgleichsgerade
np.corrcoef(x, y)
```

Balkendiagramme

```
b = plt.bar(namen, werte, color='C2', edgecolor='black')
plt.bar_label(b)                                Werte an Balken
plt.barh(namen, werte)                          waagrecht
plt.bar(x - 0.2, a, width=0.4)                  gruppiert
plt.bar(x, b, bottom=a)  plt.barh(x, b, left=a) gestapelt
```

Histogramme

```
plt.hist(daten, bins=10, edgecolor='black')
density=True  alpha=0.5  rwidth=0.9
```

Kreisdiagramme

```
plt.pie(werte, labels=namen, autopct='%1.0f%%')
startangle=90  counterclock=False
explode=[0.1, 0, 0]  colors=[...]
wedgeprops=dict(width=0.4)                      Ring
pctdistance=0.8
```

Boxplots

```
plt.boxplot([a, b], tick_labels=['A', 'B'])
patch_artist=True  orientation='horizontal'
np.median(d)  np.percentile(d, [25, 75])
```

Fehlerbalken, Flächen, Stufen

```
plt.errorbar(x, y, yerr=f, fmt='o', capsize=4)
plt.bar(x, y, yerr=f, capsize=4)
plt.fill_between(x, y1, y2, alpha=0.3)
plt.fill_between(x, y, where=y > 0)
plt.step(x, y, where='post')
np.cumsum(werte)                                Summen
```

Daten aus Dateien

```
d = np.loadtxt('daten.csv', delimiter=',',
  skiprows=1, encoding='utf-8')
usecols=(0, 2)  unpack=True  dtype=str
```

3 Layout

Figure und Axes

matplotlib_layout

```
fig, ax = plt.subplots(figsize=(6, 4))
ax.plot(x, y)  ax.scatter  ax.bar  ax.hist
ax.set_title('...')  ax.set_xlabel('x')
ax.set_xlim(a, b)  ax.set_xticks([...])
ax.legend()  ax.grid(True)
ax.set_aspect('equal')  fig.suptitle('...')
```

Mehrere Diagramme

```
fig, axs = plt.subplots(2, 3, layout='constrained')
axs[0, 1].plot(x, y)
sharex=True  sharey=True
plt.close(fig)
```

Achsen-Tricks

```
ax2 = ax.twinx()                                zweite y-Achse
ax.set_yscale('log')  ax.set_xscale('log')
ax.invert_yaxis()
ax.spines['top'].set_visible(False)
ax.spines['left'].set_position('zero')
```

Stile und Farben

```
plt.style.available
with plt.style.context('ggplot'):
cmap='viridis' 'coolwarm' 'Blues' 'gray'
```

Bilder, Heatmaps, Spielfelder

```
b = ax.imshow(a, cmap='coolwarm', vmin=0, vmax=1)
fig.colorbar(b, ax=ax)
origin='lower'  extent=[x0, x1, y0, y1]
ax.text(j, i, wert, ha='center', va='center')
from matplotlib.colors import ListedColormap
ListedColormap(['white', 'black'])
```

Höhenlinien

```
X, Y = np.meshgrid(xs, ys)
cs = ax.contour(X, Y, Z, levels=10, colors='black')
ax.clabel(cs)
ax.contourf(X, Y, Z, levels=20, cmap='terrain')
```

4 Anwendungen

Funktionen und Daten

matplotlib_anwendungen

```
np.roots(p)  np.polyval(p, x)
np.polyder(p)  np.polyint(p)
np.diff(y)  y.argmax()
ax.annotate(...)  ax.fill_between(..., where=...)
ax.errorbar(x, y, yerr=f, fmt='o')
```

Geometrie mit patches

```
from matplotlib.patches import Circle, Rectangle, Polygon
ax.add_patch(Rectangle((x, y), b, h, facecolor='C0'))
ax.add_patch(Circle((x, y), r, fill=False))
ax.add_patch(Polygon(punkte, alpha=0.5))
edgecolor='black'  linewidth=2
ax.fill(xs, ys, facecolor='C1')
```